MILITARY SPECIFICATION

FILTER UNIT, FLUID, PRESSURE MXU-408/M, ABSOLUTE 5 MICRON. HYDRAULIC

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification covers a hydraulic fluid filter unit with a replaceable element for retaining particles larger than 5 microns and 96.6 percent by weight of 5 micron and smaller particles.

1.2 Classification. Type I: 3000 psi operating pressure, 30 gpm flow. Type II: 5000 psi operating pressure. 35 gpm flow.

1.2.1 Reference Number. Specification reference number for items described in this specification will be formulated as shown in paragraph 6.4.1.

2. APPLICABLE DOCUMENTS

2.1 Government Document

2.1.1 Specifications. standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

PPP-B-601	Box. Wood, Cleated-Plywood
PPP-B-621	Box. Wood. Nailed and Lock-Corner
PPP-B-636	Box Shipping. Fiberboard

Beneficial comments (recommendations. additions. deletions) and any pertinent data which may be of use in improving this document should be addressed to:WR-ALC/MMVRG. Robins AFB. GA 31098by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4330

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Militaary

MIL-A-8625	Anodic coatings, for aluminum and aluminum alloys
MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance

STANDARDS

Military

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MS3102	Connector, Receptacle, Electric, Box Mounting
MS16142	Boss, Gasket Seal Straight Thread Tube Fitting, Standard Dimensions for
MS24391	(Supersedes AN814)
MS33649	Bosses, Fluid Connection - Internal Straight Thread
MS33656	Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Air Force-Navy Aeronautical

AN814	Plug and Bleeder - Screw Thread
AND 10050	Bosses, Standard Dimensions for Gasket Seal Straight Thread

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.

2.3 Order of precedence. in the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards). the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 GENERAL

3.1.1 Qualification. Filter units and filter elements furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.1.2 Components. The filter unit shall consist of the following major components:

Description	See Requirement
Housing	3.5.1
Element	3.5.2
Differential pressure indicator	3.5.3

3.1.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected m accordance with MI L-STD-143.

3.2 Materials

3.2.1 Protective treatment. Except for filter elements, aluminum alloy parts shall be anodized in accordance with MIL-A-8625. The use of plating to meet corrosion resistance of metals referred to in paragraph 3.2.2 is prohibited.

3.2.2 Metals. Metals shall be of the non-ferrous type in order to resist corrosion due to water, hydraulic fluids, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.2.2 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-33 586.

3.2.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the filter unit may be newly fabricated from recovered materials to the maximum extent practicable. provided the filter unit produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated m the filter unit.

3.3 Design and construction. The filter unit shall be designed and constructed so that no parts will work loose in service. The filter unit shall withstand the stresses, jars, vibrations, and other conditions incident to shipping, storage, installation, and flight line service.

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3.3.1 Functional design. The filter unit shall be designed as a light weight filter for use with hydraulic fluids conforming to MIL-H-5606 with flow rates and operating pressures per paragraph 1.2.

3.3.2 Reliability. The falter unit shall be designed and constructed as specified herein and to successfully complete the tests in paragraph 4.4.3 with no failures preventing satisfactory performance.

3.3.3 Maintainability. The filter unit shall be designed and constructed as specified herein and to provide the following.

a. Minimum number of parts consistent with the reliability and performance specified herein.

b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing.

c. Permit maintenance with general-purpose tools and equipment normally available commercially. Use of special-purpose tools and equipment shall be subject to procuring activity approval.

d. Permit adjustments, servicing, replacement of parts and components, and other maintenance with minimum disturbance to other equipment parts or components (see paragraph 3.5.1.1) Parts and components shall be located for ample and rapid access, unless performance or reliability will be appreciably degraded by the accessible location.

e. Minimum number of tools required for maintenance by design practices such as reducing the number of different bolthead sizes to the practicable minimum.

3.3.4 Foolproofness. Where improper installation of an item could cause malfunctions of that item or the system in which it is installed, an unsymmetrical mounting means shall be provided. This mounting shall be designed so that the item can only be installed in its proper operating position.

3.3.5 Element installation. The filter unit shall be designed so that it will be impossible to assemble the filter unit, if the filter element is incorrectly installed.

3.4 Performance. The filter unit shall perform satisfactorily when subjected to the following:

a. Falter assembly pressure ratings are as follows:

Class	Operating Pressure	Proof Pressure	Burst Pressure
Type I	3,000 psi	7,500 psi	11,250 psi
Type 11	5,000 psi	10,000 psi	15,000 psi

b. With a clean element installed, the differential pressure drop shall not exceed 50 psi between the relet and outlet of the housing at the rated flow per paragraph 1.2 and a fluid temperature of 100° F.

c. Actuation of the differential pressure indicator at a differential pressure of 100 ± 15 psi between the relet and outlet ports of the housing.

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d. Operating temperatures from -65° to +160°F

3.4.1 Vibration The filter unit shall be capable of withstanding the vibration specified in paragraph 4.6.6.5.1.

3.5 Details of components

3.5.1 Housing. The housing shall be designed so that flow through the element will be from the outside to the inside. The flow entering the housing shall not impinge directly upon the element filter media.

3.5.1.1 Housing design. The design of the housing shall permit change of the element without removal of the inlet or outlet line

3.5.1.2 Pressure indicator. Provisions shall be incorporated in the housing for the differential pressure indicator specified in paragraph 3.5.3.

3.5.1.3 Dimensions. The inside dimensions and the relet and outlet threads of the housing shall be shown on figures 1 and 2.

3.5.1.4 Mounting.

3.5.1.4.1 Mounting-Type I. Provisions shall be incorporated on the housing, or a corrosion resistant adapter shall be provided, to permit its being mounted on a flat vertical surface with four 11/32 inch diameter holes located in a rectangular position with hole centers spaced 4 1/2 ± 164 x 10 ± 1/64 inch. The long axis of the rectangular shall parallel to the axis of the filter elements. The top pair of holes shall be 3 3/8 ± 3/16 inch below the port-to-port centerline. The port-to-port centerline shall be parallel to the flat vertical mounting surface. in this position, with the port's end of the filter upward, the outlet port shall be on the right side. The distance from the port-to-port centerline to the verticle mounting surface shall be 3 1/2 ± 1/8 inch.

3.5.1.4.2 Mounting-Type II. Provisions shall be incorporated in the housing, or a corrosion-resistant adaptor shall be provided, to permit its being mounted on a flat vertical surface with four 11/32 inch diameter holes located in a rectangular position with hole centers spaced 6 1/2 ± 1/64 x 6 1/2 ± 1/64 inch. The top pair of holes shall be 1/2 ± 1/8 inch above the port-to-port centerline. The port-to-port centerline shall be parallel to the flat vertical mounting surface, in this position, with the port's end of the falter upward, the outlet port shall be on the right side. The distance from the port-to-port center-line to the vertical mounting surface shall be 3 1/2 ± 1/8 inch.

3.5.1.5 Wrench pad and drain port. The bottom of the housing shall have a 2 1/4 inch minimum wrench pad, and shall incorporate an MS33649-4 dram port with an AN814-4 or MS24391 plug.

3.5.1.6 Bleed port. The top of the housing shall incorporate an MS33649-4 air bleed port, located such that optimum bleeding of entrapped air within the filter will be insured. The bleed port shall be fitted with an AN814-4 or MS24391 plug.

3.5.2 Element

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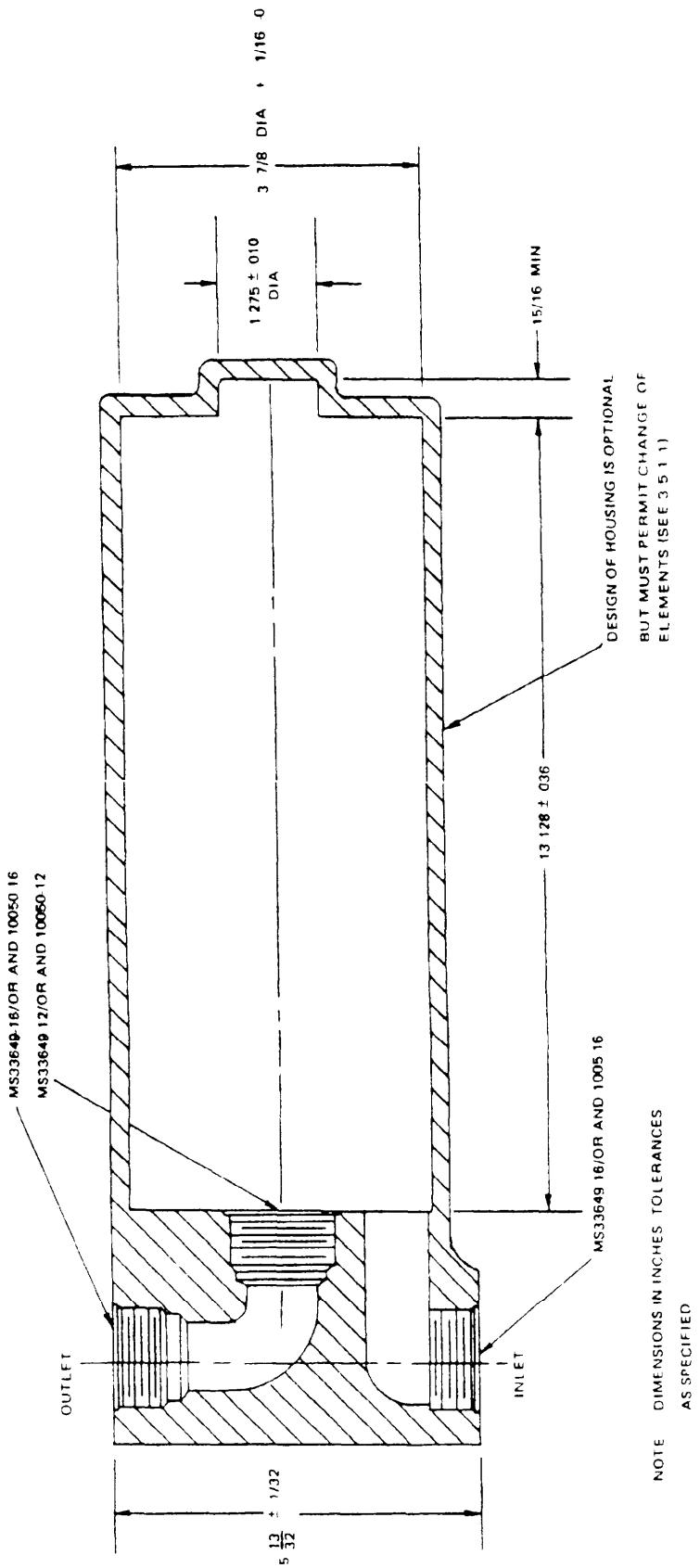


Figure 1. Dimensions of housing (Type I).

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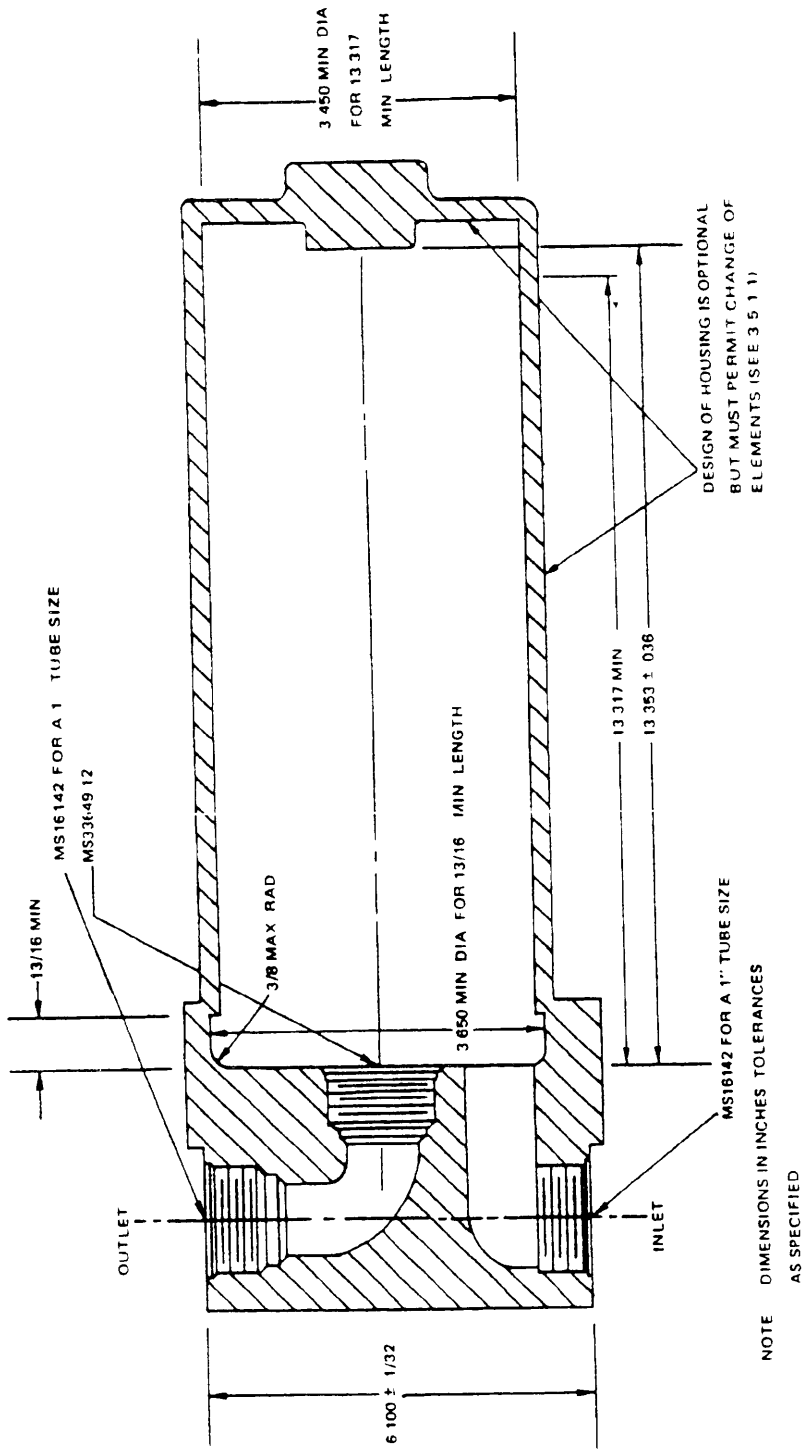


Figure 2. Dimensions of housing (Type II)

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3.5.2.1 Media. The filtering media shall be included in a self-contained cleanable and/or throw-away element with dimension envelope conforming to figure 3 for Type I, and figure 4 for Type II. The element may consist of one or two parts, with the parts being either of a cleanable or throw away type in any combination. Applicable packing shall be furnished with the elements for its installation in the housing.

3.5.2.2 Particle size. The element shall remove 96.6% by weight of all particles 5 microns and smaller in size. and shall remove 100% of all particles over 5 microns from the fluid at pressures of 0 to 3000 psi. Type I assemblies an 0 to 5000 psi. Type II assemblies, with a fluid temperature at + 50° to + 160°F,

3.5.2.3 Element material. The element shall be of a material that will not change the composition of, or be affected by, hydraulic fluids conforming to MIL-H-5606

3.5.2.4 Element performance. The element shall be of a material to maintain filtering quality and shall be supported to withstand differential pressures as follows.

a. The maximum particle size passed whale withstanding 0 to 150 psi differential pressure shall be 5 microns.

b. The maximum particle size passed whale withstanding 150 to 5,000 psi differential pressure shall be 18 microns.

3.5.2.5 Element life. Cleanable elements or parts shall withstand 1.000 hours of operation at 160° F. Noncleanable (throwaway) elements or parts shall withstand 100 hours of operation at 160°F.

3.5.2.6 Cleanable clement materials. Cleanable elements shall be of materials which will not be adversely affected by the cleaning procedures specified in paragraph 4.6.6.4f.

3.5.2.7 Dirt-holding capacity. The element shall retain a minimum of 40 grams of AC fine test dust or 200 grams of carbonyl iron E with a differential pressure not exceeding 115 psi, a fluid temperature of 100° F, and a flow of 30 gpm.

3.5.2.8 Handling damage. If applicable, the element shall be provided with a perforated cover or other device to minimize element handling damage.

3.5.3 Differential pressure indicator. The filter unit shall incorporate an integral device capable of actuating a 24-V, dc or 110-V, ac electrical signal when the differential pressure across the unit exceeds 100 ± 15 psi. Once actuated, the signal shall remain energized until the device is manually reset. The manual reset device shall be external and easily accessible. The indicator shall incorporate an MS3102R10SL-3P electrical receptacle with pin arrangement as follows:

Pin A	-	Normally open
Pin B	-	Common
Pin C	-	Normally closed

3.6 Interchangeable components. All falter units and filter elements listed on the qualified products list for each specific classification (paragraph 1.2) shall be functionally and dimensionally interchangeable.

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3.6.1 Part numbering of Interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally Interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer’s part numbers and changes thereto.

3.6.2 Weight. The dry weight of the filter unit shall not exceed 55 pounds.

3.7 Operational markings.

3.7.1 Direction of flow. The direction of flow shall be clearly and permanently indicated by at least two arrows marked on opposite sides of the housing or ports, or the ports shall be clearly marked INLET and OUTLET.

3.7.2 Instruction placard. Each filter unit shall be provided with a placard which car be permanent) mounted The placard shall describe installation and maintenance procedures and shall show an illustrated part breakdown.

3.7.3 Identification of product. The filter unit shall be marked for identification m accordance with MIL-STD-130 and as follows.

a. Manufacturer’s part number. The manufacturer’s part number and drawing number shall be the same.

b. Housing Nameplate. The housing nameplate shall be permanently marked with the following information:

Fluid Specification: MIL-H-5606
Maximum Flow: 30 gpm (Type I) or 35 gpm (Type II)
Maximum Pressure: 3000 psi (Type I)
or 5000 psi (Type II)
Procurement Data:

3.7.4 Replaceable element. The replaceable element shall be permanently marked with the following information. If marked on a sealing surface, the markings shall not affect the sealing of the element.

Manufacturer’s part number:

Manufacturer’s name or trademark:

Cleanable or noncleanable (as applicable):

Element rating: 5 micron absolute

Fluid Specification: MIL-H-5606
Procurement Data.

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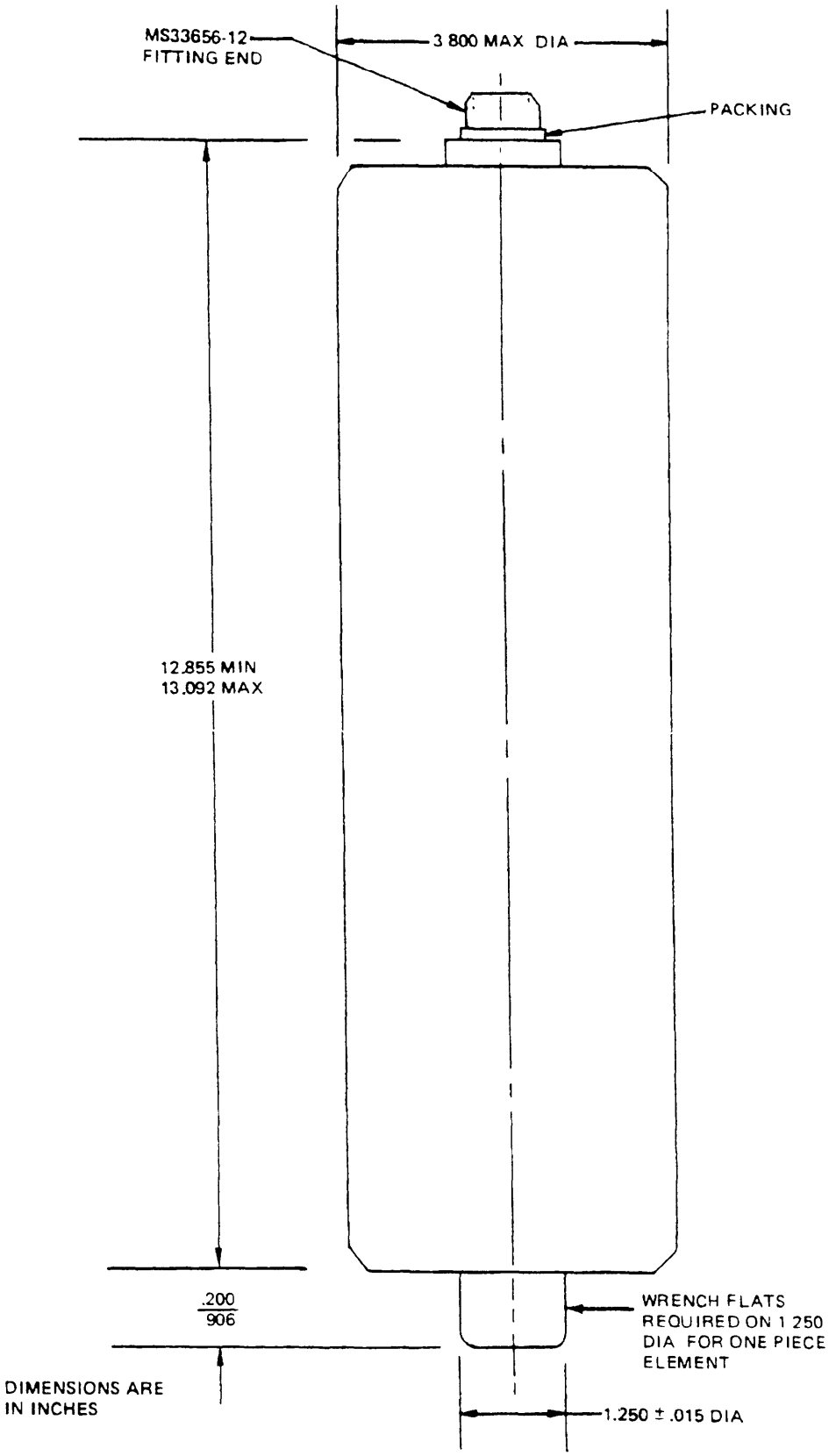


Figure 3 Element envelope dimensions (Type I).

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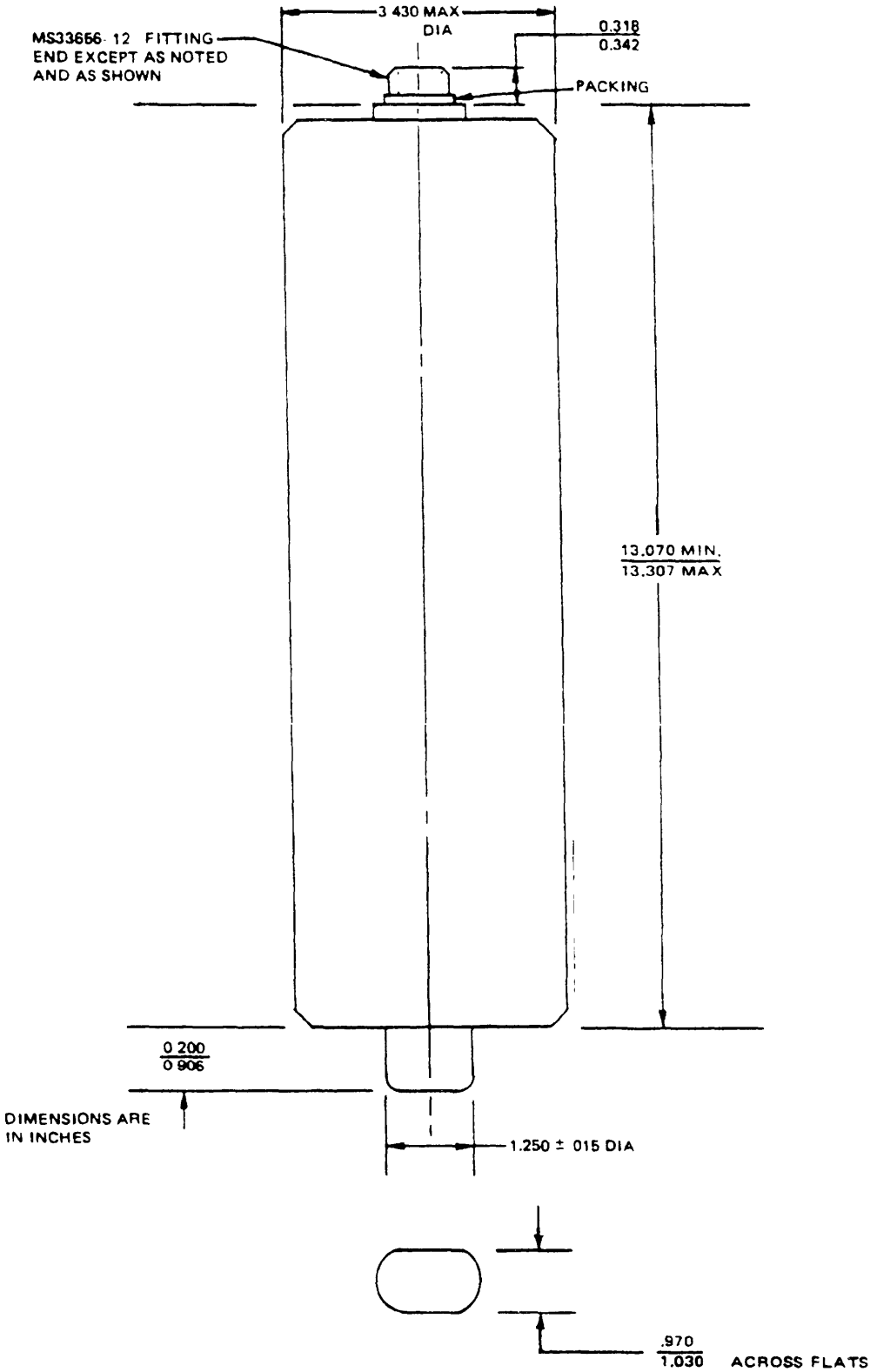


Figure 4 Element envelope dimensions (Type II)

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3.8 Workmanship. The filter unit, including all parts and assemblies, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and assemblies, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order. the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract or order, the supplier amy use his own or any other facilities suitable for the performance of the inspection requirements specified herein. unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

41.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements m specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the filter unit shall be classified as follows:

a. Qualification Tests . . . . . See paragraph 4.4

b. Acceptance tests . . . . . See paragraph 4.5

4.3 Test conditions

4.3.1 Test fluid. The hydraulic fluid used for all tests shall conform to MIL-H-5606.

4 3 2 Fluid temperature. Unless otherwise specified herein. the tests shall be conducted with the fluid at 100°F. The oil temperature during each test specified herein shall be noted in the test report. Unless otherwise specified herein, all temperatures shall be held within ± 10°F.

4.3.3 Ambient conditions. Unless otherwise specified herein, tests shall be conducted at standard atmospheric conditions (approximtaely 68° F and 29.92 inches Hg).

4,4 Qualification tests

4.4,1 Test samples. One filter unit and five elements, or five sets of element parts, shall be subjected to the qualification tests specified in paragraph 4.4.3.

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4.4.1.1 Element qualification. It is permissible to qualify the element only in this case, five sets of elements shall be subjected to tests as specified in paragraph 4.4.3 with the omission of tests in paragraphs 4.6.2 and 4.6.4.

4.4.2 Test report. After completion of the qualification tests, a test report shall be prepared m accordance with MIL-STD-831 and three complete copies of the report furnished to the procuring activity.

4.4.2.1 Reliability and maintainability information. The following information shall be submitted as an attachment accompanying the qualification test report (same quantity and quality of copies) or shall be included as part of that report.

a. All failures, maintenance, and other events recorded shall be identified by accumulated operating time. males, cycles. or position m the test procedure as appropriate. Test conditions during failures or irregular operations shall be recorded.

b. Description of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations.

c. Description of the engineering reasoning behind any corrections made, to be made on production items, or proposed to be made and behind the predicted effectiveness of those corrections.

d. Test activity or contractor comments on field conditions or procedures to be avoided or cultivated to increase the responsibility and useful life of the item.

e. Estimates (± 5 percent of actual experience) of manhours required for each maintenance and servicing action during the tests. A brief description of the qualifications and experience of the personnel revolved shall be included and shall be adequate to permit comparison to the personnel anticipated in similar field work.

f. Summary of the above information and of the results and contractor or test activity recommendations for modifications of equipment requirements, tests, or features that should improve future equipment.

4.4.3 Qualification tests. The qualification tests for the filter unit shall consist of all the tests described under paragraph 4.6. The qualification test for the element alone shall consist of the same tests described under paragraph 4.6 with the omission of paragraphs 4.6.2 and 4. 6.4. The filter housing and differential pressure indicator utilized m these element qualification tests shall be components of a falter unit which has met the qualification tests. All qualification tests shall be conducted in the order listed below.

Filter Unit	Paragraph
a. Examination of product	4.6.1
b. Proof pressure test	4.6.2

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c. Pressure drop test	4.6.3
d. Differential pressure indicator operation test	4.6.4
e. Extreme temperature operation test	4.6.5
f. Burst pressure test	4.6.9

Elements

Element No 1	Paragraph
a. Examination of product	4.6.1
b. Bubble test	4.6.6.1
c. Degree of filtration test	4.6.6.2
d Pressure impulse tests	4.6.7
e. Bubble test	4.6.6.1
f. Flow fatigue test	4.6.8
g. Bubble test	4.6.6.1

Element No. 2	
a. Examination of product	4.6.1
b. Bubble test	4.6.6.1
c. Pressure buildup and collapse (test dust)	4.6.6.4
d. Bubble test	4.6.6.1

Element No. 3	
a. Examination of product	4.6.1
b. Bubble test	4.6.6.1
c. Largest particle passed test	4.6.6.3
d. Bubble test	4.6.6.1

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Elements	
e. Pressure buildup and collapse (iron E)	4.6.6.4
f. Bubble test	4.6.6.1
Element No. 4	
a. Examination of product	4.6.1
b. Bubble test	4.6.6.1
c. Media migration and vibration tests	4.6.6.5
d. Bubble test	4.6.6.1
Element No. 5	
a. Bubble test	4.6.6.1
b. Fluid composition test	4.6.10
c. Bubble test	4.6.6.1

4.5 Acceptance test

4.5.1 Housing test. Each housing shall be subjected to the following tests:

a. Examination of product	(See paragraph 4.6.1)
b. Proof pressure test	(See paragraph 4.6.2)
c. Two cycles of the differential pressure indicator operation test	(See paragraph 4.6.4)

4.5.2 Element tests. Each element shall be subjected to the following tests:

a. Examination of product	(See paragraph 4.6.1)
b. Bubble test	(See paragraph 4.6.6.1)

4.6 Test methods

4.6.1 Examination of product. The filter unit shall be examined to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions. and marking.

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4.6.2 Proof pressure test. The filter unit shall be assembled with the outlet port plugged. The filter unit shall then be filled with clean test fluid and a pressure of 7,500 psig shall be applied for the Type I assembly and 10,000 psig for the Type II assemblies, at least twice and held for 2 minutes at each application. The pressure shall be reduced to zero between applications. Any evidence of permanent deformation malfunction, or leakage shall be cause for rejection.

4.6.3 Pressure drop test. The plug in the outlet end of the filter unit shall be removed and a pressure indicator installed at the outlet and inlet of the housing. The indicator shall be accurate to within ± 1/2 percent of full scale-reading. Elements which have successfully withstood the tests specified in paragraphs 4.6.1 and 4.6.2.1 shall be installed and a flow of 30 gpm for the Type I assemblies and 35 gpm for the Type II assemblies with an oil temperature stabilized at 100° F shall be directed through the unit The differential pressure from the inlet to the outlet shall not exceed 50 psi

4.6.4 Differential pressure indicator operation test. With the outlet of the filter unit partially blocked. the pressure at the relet to the filter unit shall be slowly increased until the differential pressure indicator is fully actuated. For this test, a 24-V or 110-V power supply and test light shall be wired through the pressure indicator device. The test light shall illuminate when the differential pressure between the inlet and outlet reaches 100 ± 15 psi. The pressure shall be raised to 5,000 psi (the outlet may be totally blocked for this portion of the test). The pressure shall be returned to zero, and the test light shall remain illuminated. The indicator shall be manually reset. The above shall be considered one complete cycle and shall be conducted 24 tunes. Twelve cycles shall be accomplished using 24-V input, and twelve cycles shall be accomplished using 110-V input. Failure of the indicator to actuate at 100 ± 15 psi on any one of the above 24 cycles shall be cause for rejection.

4.6.5 Extreme temperature operation test The filter unit shall be subjected to -65° F for 24 hours. During this period, pressure applied shall not exceed 10 psi. At the end of this period, a differential pressure of 100 ± 15 psi shall be created by applying flow through the unit and continuing this flow for 10 minutes. The differential pressure indicator shall actuate within its specified range. The flow during this test shall be recorded. Proof pressure shall then be applied to the relet port for 2 minutes with the outlet blocked. The fluid used for these tests shall be not warmer than -65° F. This test shall then be repeated at -40° and +20° F. The unit shall then be removed from the cold chamber and placed in a temperature environment of + 160° F within 20 minutes. Flow shall be resumed until a differential pressure of 100 ± 15 psi has been obtained The differential pressure indicator shall actuate within its specified range. The element shall be removed and shall successfully withstand the test specified in paragraph 4.6.6.1.

4.6.6 Element tests

4.6.6.1 Bubble test. The test element shall be installed in a setup similar to that shown on figure 5 Should the design of the filter unit make it impractical to conduct the bubble test with airflow from inside-out, the test may be conducted with airflow from outside-in provided the test setup is approved by the procuring activity. The fluid level shall be maintained 1/2 inch above the test element. The element shall be slowly rotated whale the air pressure is gradually increased by means of the needle valve. The area of greatest porosity shall be determined by observing the first bubble on the surface of the element and the manometer reading in inches of water at which this bubble emerges from the element. This shall be recorded as the initial bubble point. The initial bubble point pressure for each element shall be not less than 0.90 times the minimum qualification bubble point.

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4.6.6.2 Degree of filtration test. The hydraulic fluid and air used in the degree of filtration test shall be clean and filtered prior to the test. Figure 6 shows a degree of filtration test setup with a cleanup device attached.

4.6.6.2.1 Test procedure for determination of oil cleanliness. The test procedure for determination of oil cleanness shall be as follows.

a. A cleanup apparatus consisting of needle valves, a Millipore fixture, Millipore Type HA disc (1-inch diameter) manometer, flowmeter, and pump with adjustable internal bypass shall be installed on the test stand similar to the setup shown on figure 6.

b. A constant flow shall be established and maintained for 30 minutes, with at least periodic agitation of the contaminant mixing chamber, at a pressure differential of 50 to 60 psi across the Millipore filter assembly.

c. With the flow held constant, the pressure drop across the Millipore fixture shall be recorded at the beginning and end of the flow period.

d. The pressure drop recorded shall not exceed 1-1/2 percent of the initial value. If the variation exceeds 1-1/2 percent, the system shall be recleaned until the 1-1/2 percent maximum variation is achieved.

e. Two thousand milliliters of the clean fluid shall be collected in a 5,000-ml beaker from the sampling valve downstream of the test filter housing.

f. A membrane filter 47-mm diameter disk, absolute 1.0 micron (type AA Millipore, or equal), shall be washed with 200 ml of petroleum ether and dried to constant weight at 125° F. The weight shall be accurate to 0.1 mg. The disk shall be heated at 125°F for 30 minutes, then cooled 30 minutes in a desiccator.

g. The weighed filter membrane shall be assembled into the filter holder assembly (see figure 7) and fitted to a 5,000-ml vacuum flask connected to a vacuum pump (minimum 25 inches HG vacuum).

h. The 2,000-ml fluid sample shall be filtered through the membrane disk The beaker shall be washed with 800 ml of petroleum naptha and 800 ml of petroleum either (boiling point 95°to 130° F). The wash fluid shall be filtered through the membrane disk. One hundred milliliters of petroleum naphtha and one hundred milliliters of petroleum ether shall then be used to wash down the funnel, and then this wash shall be filtered.

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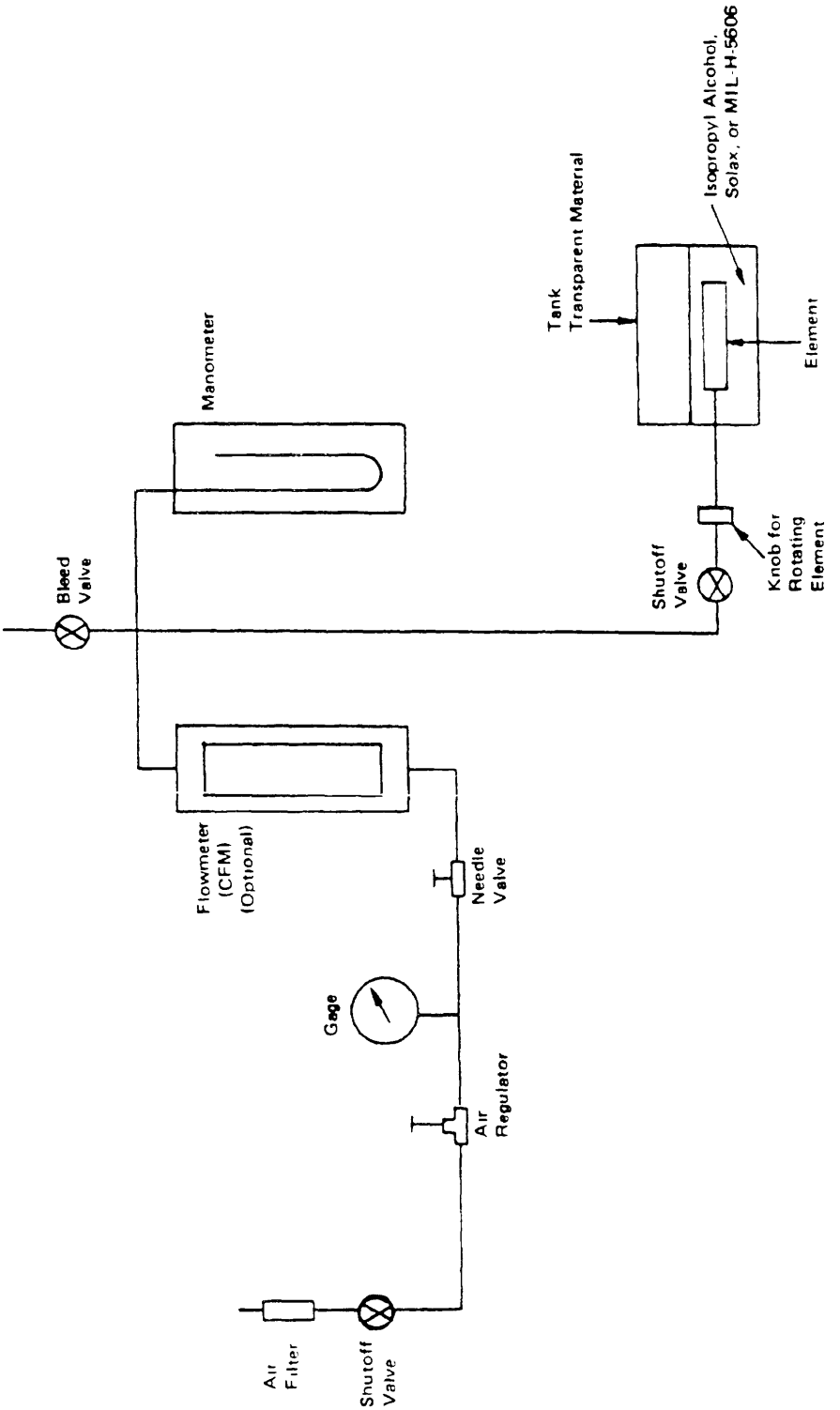


Figure 5 Typical schematic diagram for air bubble test

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i. The membrane disk shall then be removed and dried to a constant weight at 125°F and weighed to an accuracy of 0.1 mg The contaminant weight shall be less than 0.0004 gram/2,000 ml.

4.6.6.2.2 Filter element tests. The degree of filtration for filter elements shall be determined as follows

a . The system shall be checked for oil cleanliness in accordance with paragraph 4.6.6.2.1

b . Prefiltered hydraulic fluid shall be added through plug valve A to bring the level in the mixing chamber to 2,000 ml. The element shall be installed in the filter housing.

c. Valve B shall be closed.

d. A 10-ml slurry containing 3 grams of contaminant shall be added to the fluid through a small funnel inserted in plug valve A. The contaminant shall be 1.5 grams of APM F-9 beads (see figure 8) and 1.5 grams of standardized fine air cleaner (AC) test dust (see paragraph 4.6.6.4d).

e. The contaminant shall be uniformly distributed by churning the fluid for 5 minutes with an agitator.

f. Plug valve A shall be closed, and the glass chamber containing hydraulic fluid and contaminant shall be pressured by the air regulator. The air regulator shall be used to maintain flow. Sufficient air pressure shall be employed to maintain rated flow.

g. Gate valve B shall be opened and air pressure shall torte fluid containing contaminant through the sample filter unit at rated flow. This 2,000 ml of filtrate shall be collected m a 5,000-ml beaker from sampling valve C. Seven hundred and fifty milliliters of petroleum ether (boding point 95° to 130° F) shall then be washed through the contaminant mixing chamber and test falter unit using wash bottle. The wash fluid shall be collected in the same 5,000-ml beaker.

h. A Type AA4-047 Millipore membrane falter paper disk shall be washed with 200 ml of petroleum ether and dried to constant weight at 125° F. The weight shall be accurate to 0.1 mg.

i. The weight filter membrane shall be assembled into the filter holder assembly (see figure 6) and fitted to a 5,000-ml vacuum flask connected to a vacuum pump (minimum 25 inches Hg vacuum).

j. All fluid passed through the test filter shall be filtered through the membrane disk. The 5,000-ml beaker shall then be washed with 500 ml of petroleum ether (boding point 95° to 130° F). and this fluid from the washed beaker shall pass through the membrane. The falter funnel shall be washed with 100 ml of petroleum ether and this wash passed through the membrane.

k. Maintaining vacuum. the top half of filter funnel assembly shall be removed, exposing the filter paper for further washing.

l. With a wash bottle of faltered petroleum ether (boiling point 95° to 130° F), the rim of the filter paper shall be gently washed to remove the traces of oil, being careful not to disturb the contaminant cake. Vacuum shall be maintained during the above operation.

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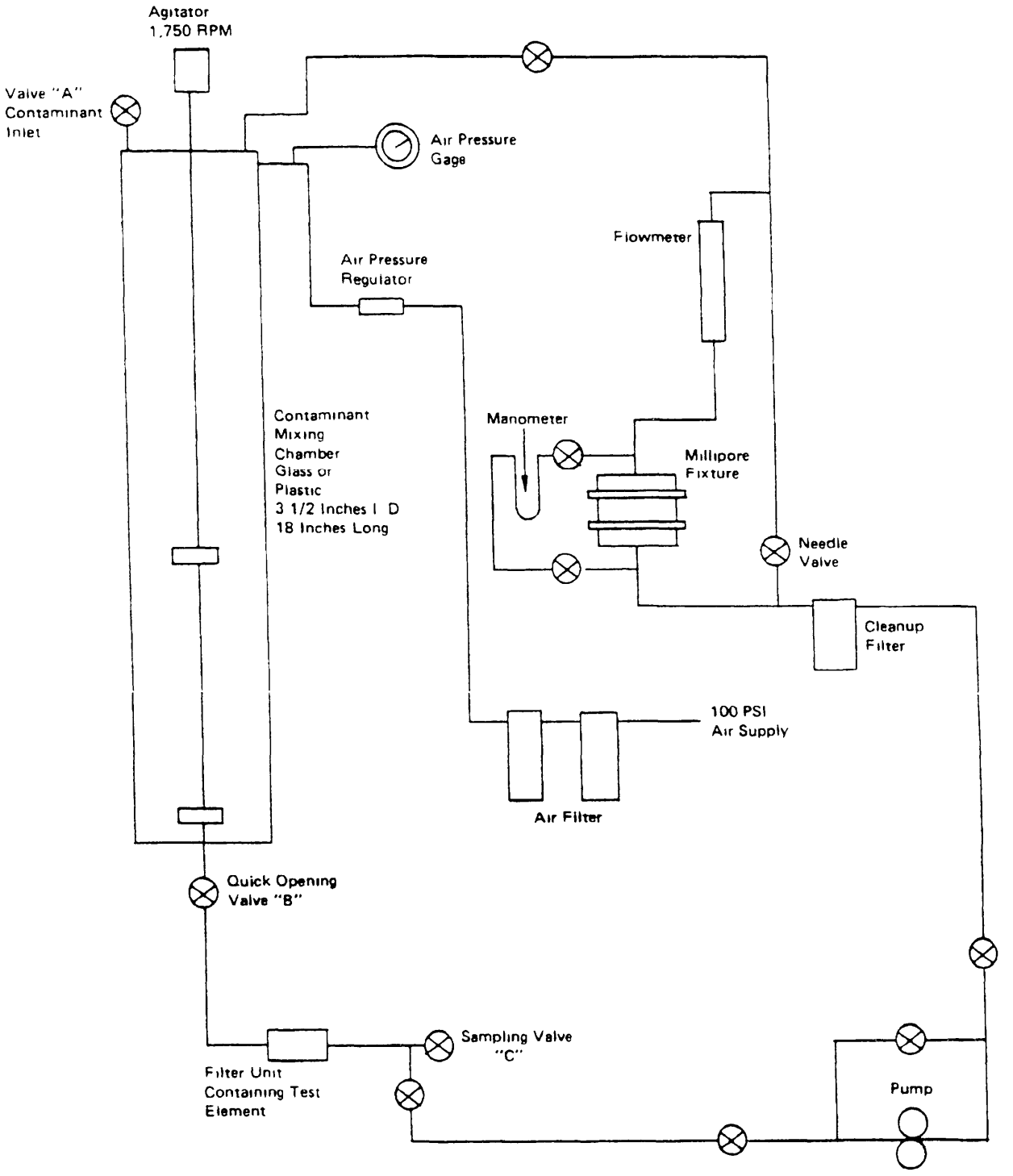


Figure 6. Typical apparatus for determination of efficiency of filter element.

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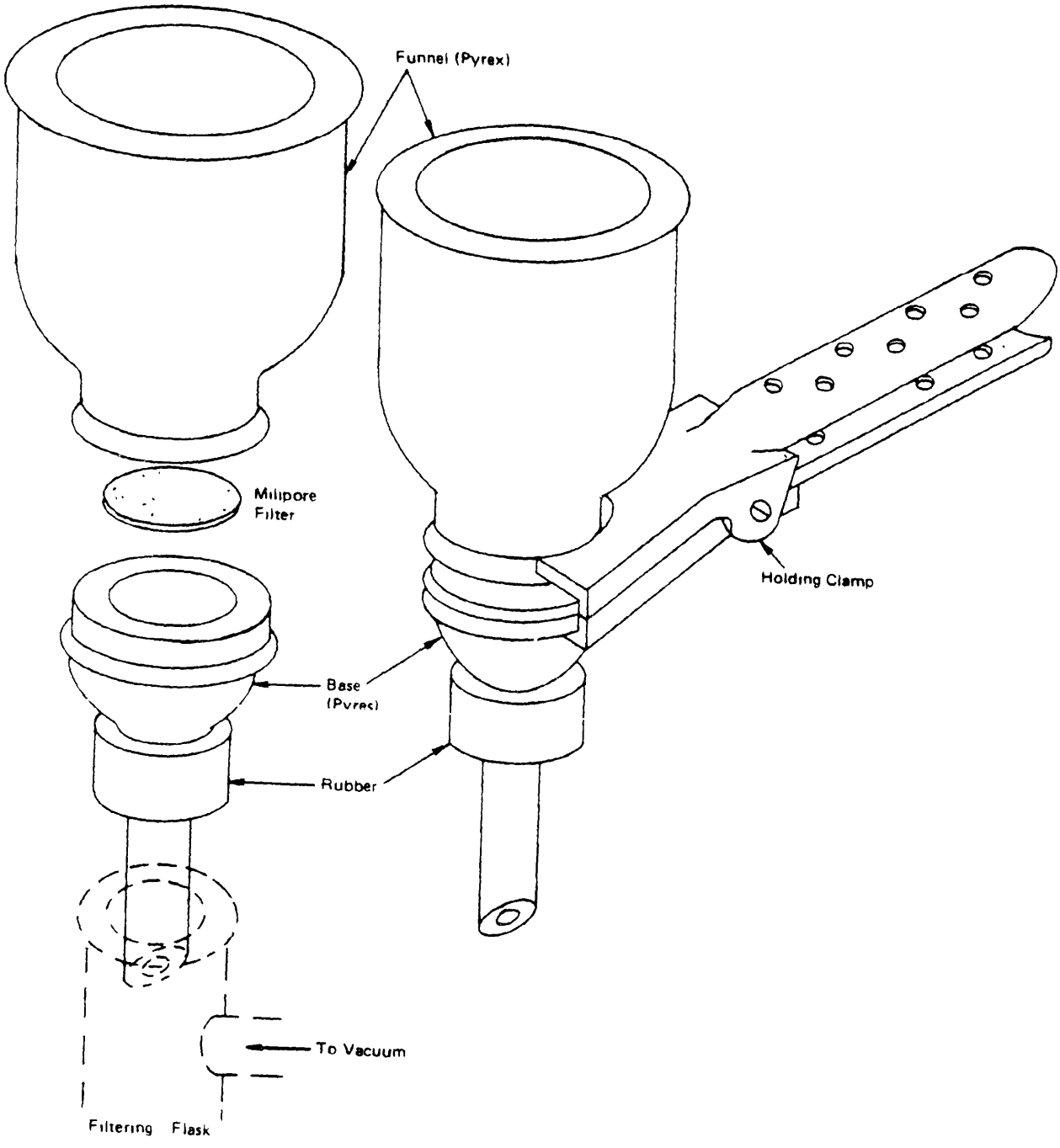


Figure 7 Diagram of effluent filtration setup

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m. The vacuum shall be shut off and the filter paper removed. It shall be dried to a constant weight at 125°F and weighed to an accuracy of a tenth of a milligram. The difference between the filter paper weights before and after the above procedure shall be the weight of contaminant passing through the test filter B in the formula below. The above test, with the element removed, shall be repeated four times for tour separate contaminant add values (A in formula below). Oil cleaned to a level that conforms to paragraph 4.6.6.2 .1 shall be used for each test. The contaminant collected shall be the acid value and shall be used in the calculation of the degree of efficiency, It shall be the average of the tour runs. A system blank value shall be obtained by repeating steps a through m above with the falter element installed and no contaiminant added This blank value shall be the value C in the following efficiency formula and shall be less than 0.0007 gram:

$$\% \text{ removal} = \frac{A - (B-C)}{A} \times 100$$

No one of the four add values shall be less than 95 percent by weight of the contaminant introduced.

n. The falter unit shall remove 99.3 percent by weight of the contaminant add value This is equivalent to 100-percent removal of particles over 5 microns and 96.6 percent removal of particles 5 microns and smaller in size, by weight.

4.6.6.5 Largest particle passed test. A filtration test shall be conducted as follows

a. A degree of filtration test shall be conducted m accordance with the procedure specified in paragraph 4.6.6.2.2 a through e with 2,000 ml of hydraulic fluid. The combined added contaminant shall be 1.5 grams of APM F-9 beads and 1.5 grams of carbonyl iron E.

b. The effluent shall be passed through art absolute 1.0- micron membrane disk Type AA Millipore filter paper, or equal.

c. The entire disk shall be scanned. The largest bead shall be no greater than 5 microns

4.6.6.4 Pressure buildup and collapse test. A pressure buildup and collapse test shall be conducted as follows.

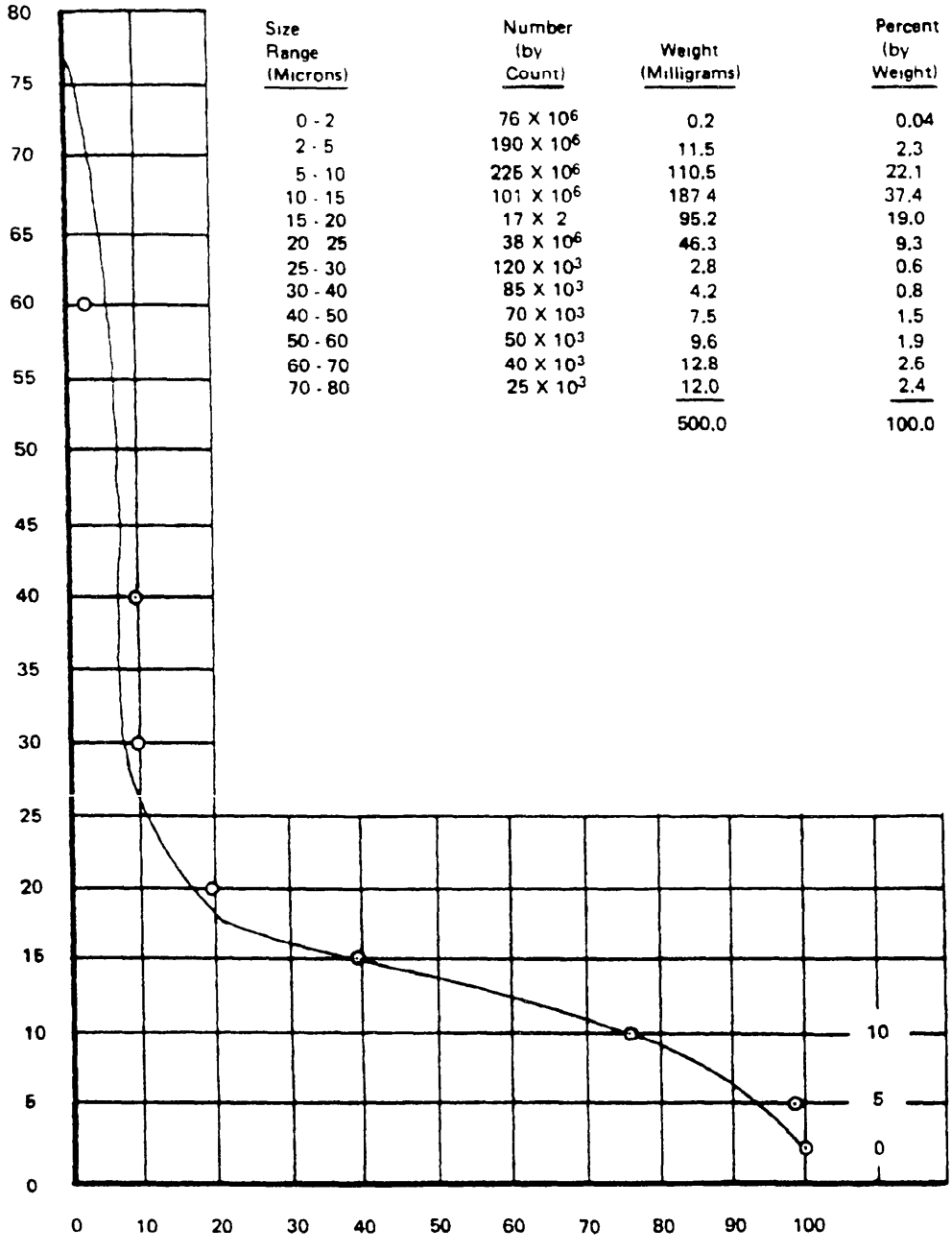
a The test fluid used in the system shown on figure 9 shall be precleaned by flowing through a cleanup filter using the test procedure specified m paragraph 4.6.6.2.1. The cleanup filter shall be bypassed throughout the remainder of this test

b. The element under test shall be installed in the housing.

c. The pump shall be started, valves A and B opened, and rated flow attained by adjusting the variable speed drive.

d. Standardized fine air cleaner (AC) test dust or carbonyl iron E shall be added m increments of 4 grams at 4-minute intervals through the dust valve, in slurry. Test dust shall have the following composition:

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Size Range (Microns)	Number (by Count)	Weight (Milligrams)	Percent (by Weight)
0 - 2	76 X 10^6	0.2	0.04
2 - 5	190 X 10^6	11.5	2.3
5 - 10	225 X 10^6	110.5	22.1
10 - 15	101 X 10^6	187 4	37.4
15 - 20	17 X 2	95.2	19.0
20 25	38 X 10^6	46.3	9.3
25 - 30	120 X 10^3	2.8	0.6
30 - 40	85 X 10^3	4.2	0.8
40 - 50	70 X 10^3	7.5	1.5
50 - 60	50 X 10^3	9.6	1.9
60 - 70	40 X 10^3	12.8	2.6
70 - 80	25 X 10^3	12.0	2.4
		500.0	100.0

Cumulative Weight Fraction – Percent Points on the Line Indicate Portions of Particles with Sizes Greater Than Diameter Noted.

Figure 8. Particle size distribution curve APM F-9 glass beads.

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Size of Particle (Micron)	Percent by Weight of Total
0 to 5	39 ± 2
	18 ± 3
10 to 20	16 ± 3
20 to 40	18 ± 3
Over 40	9 ± 3

e. Two minutes after each dust addition, pressure differential, flow and temperature shall be recorded For element No. 2 (see paragraph 4.4.3), AC fine test dust shall be added until 40 grams of dust have been added. For element No. 3 (see paragraph 4.4.3), 200 grams of carbonyl iron E shall be added. The differential pressure across the filter unit in each test shall not exceed 115 psi. The pump shall not be stopped during any test. Then, for both element No. 2 and No. 3, additional AC fine test dust, carbonyl iron E (1-12 micron) and glass beads shall be added m equal proportions until a differential pressure of 5,000 psi is obtained across the element. The dirt capacity of 5.000 psi shall be recorded. The effluent shall be sampled and tested m accordance with paragraph 4.6.6.3a and b as the differential reaches 100, 150, 1000, 2000, 3000 and 5000 psi. The samples collected at 100 and 150 psi differential shall show a maximum particle size of 5 microns. The samples collected at 1000-, 2000-, 3000- and 5000-psi differential shall show a maximum particle size of 18 microns.

f. Cleanable elements shall be loaded 10 tries to then rated differential pressure and cleaned 10 times. After the final cleaning, the dirt-holding capacity shall be not less than 90 percent of the original dirt-holdmg capacity at the elements rated differential pressure. Elements shall be cleaned by chemicals and ultrasonic equipment in accordance with recommendations submitted by the manufacturer.

g. Cleanable elements shall be subjected to the test specified in 4.6.6.1.

4.6.6.5 Media migration and vibration tests. The filter unit shall be mounted on a test stand and the oil in the test system shall be precleaned using the procedure specified in paragraph 4.6.6.2.1 The filter unit. filled to the top of the element with fluid, shall be maintained at 160°F for 72 hours.

4.6.5.1 Vibration test. The filter unit shall be cooled to room temperature (70° F to 90° F) and vibrated in three mutually perpendicular planes. The frequency of vibration shall be varied from 50 to 500 Hz. with one pass of scanning to be at least 15 minutes, at an acceleration of 10g m each plane with any resonating frequencies noted. The filter unit shall be vibrated for 1 hour at each of these resonating frequencies. If no resonance is encountered, the vibrating frequency shall be 500 Hz for 1 hour at an acceleration of 10g in each plane. The differential pressure indicator shall not actuate during this test.

4.6.6.5.2 Media migration test. At the completion of the vibration period and with the test pumping system and valves preset to give rated flow through the filter, the system shall be started and 1 gallon of fluid passed through the filter and effluent collected. The sample effluent shall be completely filtered through type AA Millipore paper (47-mm disk) and the amount collected and scanned under a

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microscope. There shall be no migration traceable to the element or its packaging. After this test, the filter shall successfully withstand the test specified in paragraph 4.6.6.1.

4.6.7. Pressure inpulse tests.

4.6.7.1 Filter housing test. The falter housing shall be subjected to 100,000 impulse cycles at 160° F. Each impulse cycle shall consist of a pressure rise from operating pressure (Type I), or a pressure less than 100 psi (Type II), to a peak surge pressure of 1.331 ± 0 05 tunes the rated operating pressure, as shown by an oscilloscope. The impulse time (duration of impulse) shall not be longer than 0.05 seconds and the rate of pressure rise shall be a minimum of 50,000 psi per second. Cycling shall be performed at a maximum rate of 300 cycles per minute. The differential pressure indicator shall not actuate during the impulse test There shall be no evidence of external leakage or structural failure during this test.

4.6.7.2 Filter element test. The filter element impulse test shall be run simultaneously with the housing impulse test. Noncleanable elements shall withstand 10,000 impulse cycles without evidence of structural failure. Cleanable elements shall withstand 100,000 cycles without evidence of structural failure.

4.6.8 Flow fatigue test.

4.6.8.1 Initial bubble point. Elements to be submitted for flow fatigue tests shall be subjected to the test m paragraph 4.6.6.1 to establish an initial bubble point pressure.

4.6.8.2 Filter element test. Filter elements shall be installed m the housing and subjected to 3,000 psi relet pressure and rated flow with fluid at 160°F. Elements shall be loaded with AC fine test dust and glass beads to produce the specified differential pressure across the housing relet and outlet. Elements or parts of elements that are designed to withstand 0 to 150 psi differential pressure in accordance with paragraph 37.2.4a shall withstand series I tests only. Elements or parts of elements that are designed to withstand 150 to 5,000 psi differential pressure in accordance with paragraph 3.7.2.4b shall withstand series II tests only. A cycle shall consist of increasing the differential pressure across the filter unit from 0 to the values specified and back to 0 by first increasing then decreasing flow through the filter unit. The cycling rate shall not exceed 300 cycles per minute The number of cycles at each differential pressure shall be as follows

Series 1

a 3,000 cycles at 55 psi pressure drop.

b. 6,000 cycles at 75 psi pressure drop

c. 1,000 cycles at 100 psi pressure drop.

Series II

a. 40,000 cycles at 75 psi pressure drop.

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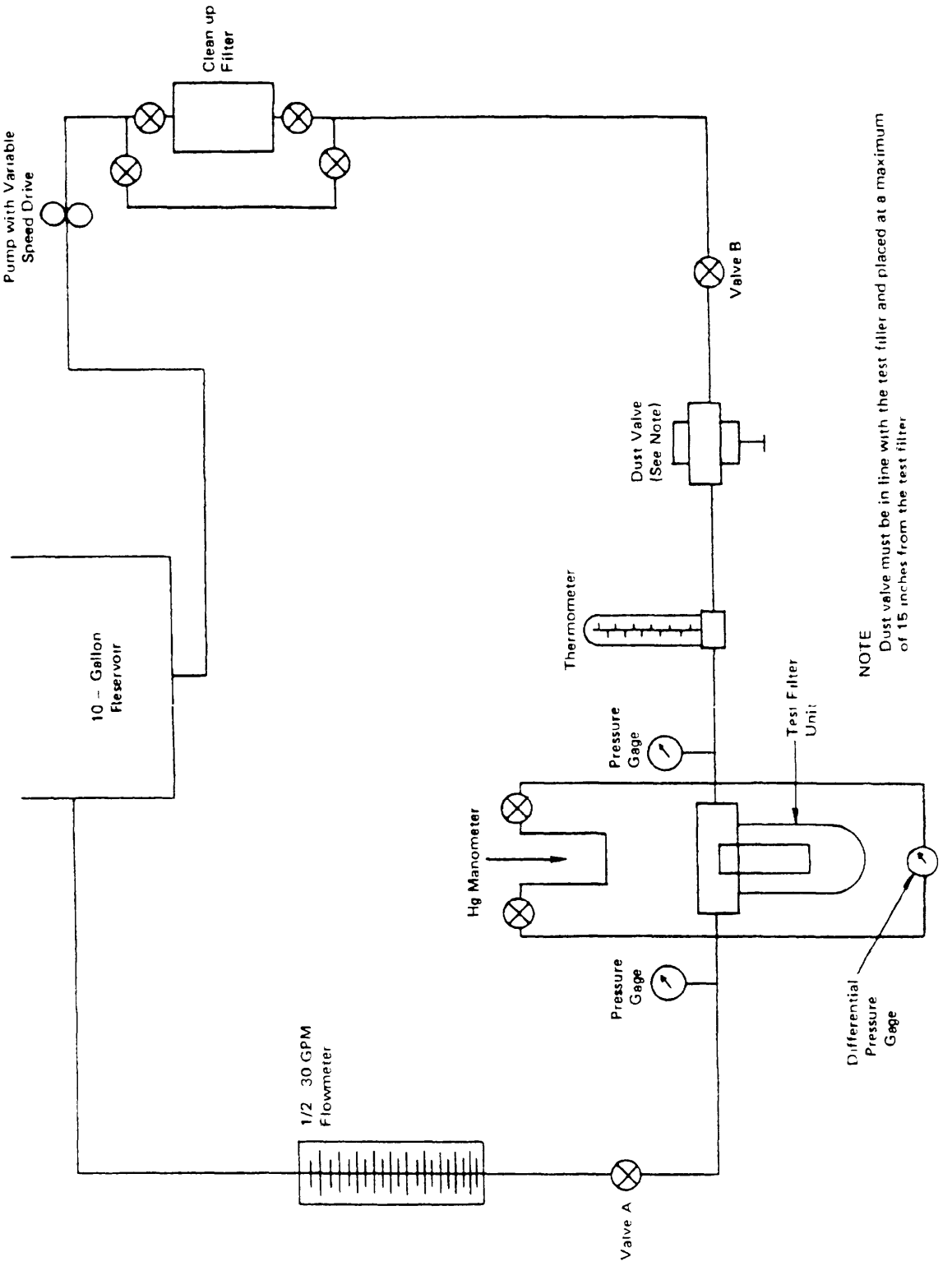


Figure 9 Typical setup for determining pressure buildup and collapse pressure characteristics

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b 60,000 cycles at 100 psi pressure drop

c. 100 cycles at 3,000 psi pressure drop.

46.8.2.1 Particle size. Near the completion of each subseries test a 2,000-ml sample of effluent fluid shall be collected. The fluid sample shall be passed through an absolute 1.0-micron membrane disk type AA Millipore filter paper. or equal. The disk shall be scanned. Any particle larger than the original micron rating of the element under test shall be cause for rejection.

4.6.8.2.2 Final bubble point. At the completion of each subseries test, elements shall be subjected to the test specified m paragraph 4.6.6.1. A decrease in bubble pressure from the initial value. as recorded during the test specified m paragraph 4.6.8.1. shall be cause for rejection.

4.6.9 Burst pressure test. The burst pressure test shall be conducted at 160° F after a five hour soak at 160° F with the falter unit filled with test fluid. Pressure shall be applied with a hand pump and a maximum rate of increase of 25,000 psi per minute until the burst pressure, as specified in paragraph 3.4a. is reached. Filter units shall show no leakage m the forms of drops or rupture of internal or external parts at this pressure when held for 2 minutes. Following this test, the pressure may be increased above that specified in order to secure data on actual burst pressure.

4.6.10 Fluid composition test. The cleanup filter shall be removed from the setup shown on figure 9. The reservoir and circuit shall be filled with unfiltered hydraulic fluid. Two 1-quart samples of the unfiltered fluid shall be collected. The fluid shall then be circulated through the circuit at 30 gpm and two 1 quart samples of the effluent taken. Test dust shall be added at the dust valve until the differential pressure across the filter unit reaches 100 psi. Two 1-quart samples of the effluent shall then be collected. One of each of the three samples of fluid shall be properly identified and shall be submitted to all the tests of MIL-H-5606. The three remaining samples shall be properly identified and forwarded to the procuring activity. Any change to the composition of the hydraulic fluid as a result of passing through the filter shall be cause for rejection.

4.7 Inspection of the preservation, packaging, and marking for shipment and storage. The inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

5.1 Packaging (see paragraph 6.2)

5.1.1 Level A. One each falter unit or element shall be packaged in accordance with method II of MIL-P-116. All openings of the filter unit or element shall be plugged or sealed with approved noncorrosive materials to prevent the entry of moisture, dirt, and foreign matter. The falter unit or element shall be wrapped m grade A barrier material conforming to MIL-B-121 and placed in a container conforming to PPP-B-636.

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5.1.2 Level C. The filter or element shall be packed in accordance with standard commercial practice.

5.2 Packing (see paragraph 6.2).

5.2.1 Insofar as practical, shipping containers shall be of uniform size and shape and of minimum cube and tare weight. Each shipping container shall contain only one filter unit. or one element when shipping only filter elements.

5.2.2 Level A. Unless otherwise specified, shipping containers shall be nailed wooden or wood-cleated plywood boxes conforming to PPP-B-621 or PPP-B-601 (overseas type).

5.2.3 Level C. Unless otherwise specified the unit package shall be packed in substantial commercial shipping containers constructed to insure acceptance by common or other carrier for sate transportation at the lowest rate to the point of delivery. Except as specified herein, the containers shall conform to the requirements of consolidated freight classification rules in effect at the time of shipment.

5.3 Shipment marking. In addition to any special marking required by the contract or order (see paragraph 6.2), all shipping containers shall be marked in accordance with MIL-STD-129.

a. The nomenclature for the filter unit shall be as follows:

FILTER UNIT, FLUID, PRESSURE MXU-408/M, ABSOLUTE 5 MICRON HYDRAULIC

b. The nomenclature for the element shall be as follows:

ELEMENT FOR FILTER UNIT, FLUID, PRESSURE MXU-408/M ABSOLUTE 5 MICRON, HYDRAULIC

6. NOTES

6.1 Intended use. The MXU-408/M filter unit covered by this specification is intended for use in the high-pressure outlet line of hydraulic test equipment.

6.2 Ordering data. Procurement documents should speedy the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of packaging and packing. (See section 5.)

c. Special shipment marking. (see paragraph 5.3)

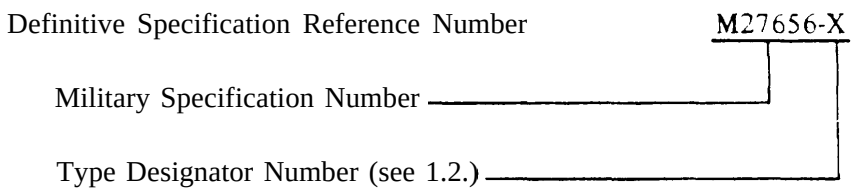
6.3 Qualified products list. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have

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the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Warner Robins Air Logistics Center/MMVRG Robbins, AFB, GA 31098, and information pertaining to qualification of products may be obtained from that activity.

6.4 International standardization. This entire specification is the subject of international agreement Air Standardization Coordinating Committee Air Standard 11/14 Certain provisions of paragraphs 1.1, 3.5.2.2, 4.6.6.2.2n, and 4.6.6.3c of this specification are the subject of international standardization agreement North Atlantic Treaty Organization Standardization Agreement No 3510. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

6.4.1 Definitive specification reference number. The reference number is a definitive reference number which will be formulated to identify each item covered by this specification. This reference number will be added to the NSN only. The reference number will be formulated by selecting from the requirements options available in this specification as follows.



6.5 Abbreviations. In order to save space and to avoid the spelling out of repetitious words and phrases, the folio wing abbreviations have been used m the text of this specification:

ac . . . . . alternating current

de . . . . . direct current

dia . . . . . diameter

g . . . . . gravitational unit

gpm . . . . . gallons per minute

Hp. . . . . mercury

Hz. . . . . Hertz, cycles per second

mg . . . . . milligram

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mm . . . . . minimum

ml . . . . . milliliter

mm . . . . . millimeter

psi. . . . . pounds per square inch

pig. . . . . pounds per square inch gage

rpm . . . . . revolutions per minute

v . . . . . volt

6.6 Subject Term (Key Word) Listing

Absolute
Fluid
Hydraulic
Pressure

6.7 Identification of changes. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:

Air Force - 99

Review activity:

Army - AV

Air Force - 82

Preparing activity:

Air Force - 84

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1. DOCUMENT NUMBER	2 DOCUMENT TITLE
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3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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